Lecture 5

Observables and the Poisson algebra

Objectives. *Lecture 4 gave Hamilton's equations. This lecture asks how an observable — anything one can measure on a state, a function on phase space — changes along a motion. The answer is its bracket with the Hamiltonian. The* Poisson bracket *makes the observables an algebra. We prove its properties, including the Jacobi identity, and express conservation laws as bracket equations. Conserved quantities then form a Lie algebra. Angular momentum gives the example* $\mathfrak{so}(3)$. *Finally, we describe time evolution by letting observables change while keeping the initial state fixed. These structures begin the Poisson layer of the course, and they are the first part of its second answer.*

By the end of this lecture you will be able to:

- Compute canonical Poisson brackets, including those of angular momentum.
- Test conservation using the bracket with the Hamiltonian.
- Use the Jacobi identity to show that conserved quantities form a Lie algebra.
- Evolve oscillator observables in the Heisenberg picture.

5.1 Observables

Throughout Part II the phase space is an open set $\mathcal{M} \subseteq \mathbb{R}^{2n}$. We now write $x = (q, p) = (q_1, \dots, q_n, p_1, \dots, p_n)$ for a point of it, called z in (4.9), and the motion is given by Hamilton's equations (4.8), $\dot{q} = \partial H/\partial p$, $\dot{p} = -\partial H/\partial q$, for a Hamiltonian $H(q, p, t)$ of class C^∞ on $\mathcal{M} \times I$, I an open interval (Theorem 4.5). The pendulum lives on $\mathbb{R}^2$, with the angle read modulo 2π (§ 1.5).

Definition 5.1 (Observable). An *observable* is a function $f \in C^\infty(\mathcal{M})$, real valued. A *time-dependent observable* is a C^∞ function $f(q, p, t)$ on $\mathcal{M} \times I$.

A state determines the value of every observable. An observable models a measuring device whose reading on the state x is $f(x)$ (Figure 5.1). Observables form a real vector space and can be multiplied pointwise, and a smooth function of observables is an observable: if $f_1, \dots, f_k \in C^\infty(\mathcal{M})$ and $F \in C^\infty(\mathbb{R}^k)$, then $F(f_1, \dots, f_k) \in C^\infty(\mathcal{M})$ is the device "read $f_1, \dots, f_k$, then compute F". So $C^\infty(\mathcal{M})$ is a commutative algebra with unit 1. The examples are the coordinates q_i and p_i, the kinetic energy $|p|^2/2m$, and, for a particle in $\mathbb{R}^3$, the angular momentum $\mathbf{L} = q \times p$, whose components are three observables.

Some natural questions are not smooth. "Is the particle in the box B?" is the indicator function $\mathbf{1}_B(q)$, which jumps on the boundary of B. It is not an observable in the sense of Definition 5.1: the bracket below differentiates, and $\mathbf{1}_B$ cannot be differentiated there. Its role is statistical, the probability of finding the particle in B, and Lecture 7 describes this probability by treating states as probability distributions. Smoothness is also why we take

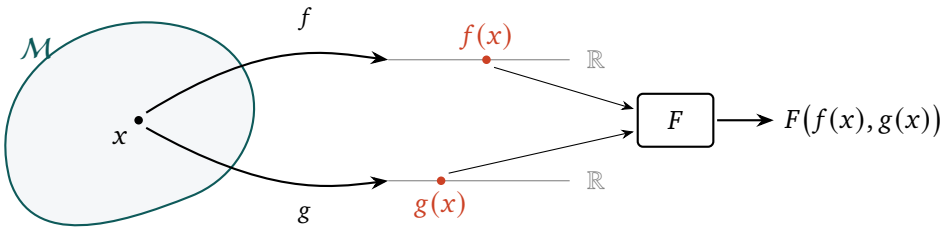


Figure 5.1: Observables as measuring devices, after Nestruev, *Smooth Manifolds and Observables*, §1. A state x gives a reading of each device; a smooth function of readings is again a device.

C^∞ rather than C^2: a bracket costs one derivative, and C^∞ is the class that brackets never leave.

5.2 The Poisson bracket

How does the reading of an observable change along a motion? Let $x(t) = (q(t), p(t))$ solve Hamilton's equations. By the chain rule,

$$\frac{\mathrm{d}}{\mathrm{d}t} f\big(x(t), t\big) = \sum_{i=1}^{n} \Big(\frac{\partial f}{\partial q_i} \dot{q}_i + \frac{\partial f}{\partial p_i} \dot{p}_i \Big) + \frac{\partial f}{\partial t} = \sum_{i=1}^{n} \Big(\frac{\partial f}{\partial q_i} \frac{\partial H}{\partial p_i} - \frac{\partial f}{\partial p_i} \frac{\partial H}{\partial q_i} \Big) + \frac{\partial f}{\partial t}.$$

The sum depends only on f and H. It is their Poisson bracket.

Definition 5.2 (Poisson bracket)**.** For $f, g \in C^\infty(\mathcal{M})$,

$$\{f, g\} = \sum_{i=1}^{n} \Big(\frac{\partial f}{\partial q_i} \frac{\partial g}{\partial p_i} - \frac{\partial f}{\partial p_i} \frac{\partial g}{\partial q_i} \Big). \tag{5.1}$$

For time-dependent observables the bracket is taken at each fixed t.

So, along every solution of Hamilton's equations,

$$\frac{\mathrm{d}}{\mathrm{d}t} f\big(x(t), t\big) = \Big(\{f, H\} + \frac{\partial f}{\partial t} \Big)\big(x(t), t\big). \tag{5.2}$$

This is the sign convention of this course.[1] In the notation of (4.9), $\{f, H\} = \nabla f \cdot J \nabla H$ is the derivative of f along the velocity field $J\nabla H$ of the motion: *it measures how fast the motion cuts across the level sets of* f, and it vanishes where the motion is tangent to them (Figure 5.2).

The coordinates themselves give the simplest brackets. Since $\partial q_i / \partial q_j = \delta_{ij}$ and $\partial q_i / \partial p_j = 0$,

$$\{q_i, q_j\} = 0, \qquad \{p_i, p_j\} = 0, \qquad \{q_i, p_j\} = \delta_{ij}, \tag{5.3}$$

the *canonical relations*, and for every observable f

$$\{f, p_i\} = \frac{\partial f}{\partial q_i}, \qquad \{f, q_i\} = -\frac{\partial f}{\partial p_i} : \tag{5.4}$$

[1] Faddeev and Yakubovskii, and Takhtajan, write $\sum (\partial_p f\, \partial_q g - \partial_q f\, \partial_p g)$, the negative of (5.1); their $\dot f = \{H, f\}$ is our $\dot f = \{f, H\}$.

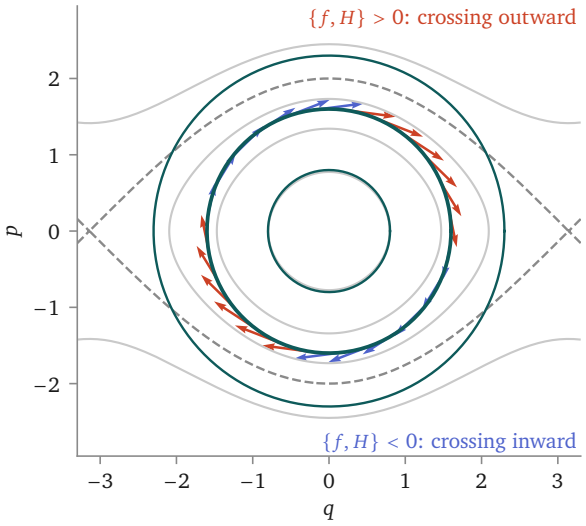


Figure 5.2: The pendulum $H = \frac{1}{2}p^2 + 1 - \cos q$ (grey level sets; dashed: the separatrix) and the circles $f = \frac{1}{2}(q^2+p^2) = \text{const}$ (teal). On one circle the arrows are the velocity field $J\nabla H = (p, -\sin q)$. Since $\{f, H\} = p(q - \sin q)$, they point outward where it is positive, inward where it is negative, and along the circle where it vanishes.

brackets with the coordinates are partial derivatives. With $f = q_i$ and $f = p_i$ in (5.2), Hamilton's equations become

$$\dot{q}_i = \{q_i, H\}\,, \qquad \dot{p}_i = \{p_i, H\}\,. \tag{5.5}$$

For the oscillator $H = p^2/2m + \frac{1}{2}m\nu^2q^2$ of Lecture 4, $\{q, H\} = p/m$ and $\{p, H\} = -m\nu^2 q$.

For the pendulum $H = \frac{1}{2}p^2 + 1 - \cos q$ and $f = \frac{1}{2}(q^2 + p^2)$,

$$\{f, H\} = \frac{\partial f}{\partial q}\frac{\partial H}{\partial p} - \frac{\partial f}{\partial p}\frac{\partial H}{\partial q} = qp - p\sin q = p\,(q - \sin q).$$

Since $q - \sin q$ has the sign of q, $\{f, H\} > 0$ exactly where $qp > 0$: in the first and third quadrants the motion crosses the circles outward, in the second and fourth inward (Figure 5.2). The distance from the origin is not conserved, as the non-circular level sets of H show.

5.3 The algebra

Proposition 5.3 (Bilinear, antisymmetric, Leibniz)**.** *For observables f, g, h, real a, b and $F \in C^\infty(\mathbb{R}^k)$:*

1. $\{f, ag + bh\} = a\,\{f, g\} + b\,\{f, h\}$;
2. $\{g, f\} = -\{f, g\}$; *in particular* $\{f, f\} = 0$;
3. *(Leibniz)* $\{f, gh\} = \{f, g\}\,h + g\,\{f, h\}$, *and more generally* $\{f, F(g_1, \dots, g_k)\} = \sum_j \partial_j F(g_1, \dots, g_k)\,\{f, g_j\}$.

Proof. (5.1) is linear in g and changes sign when f and g are exchanged. For the composition rule, the chain rule gives $\partial F(g_1, \dots, g_k)/\partial p_i = \sum_j \partial_j F\,\partial g_j/\partial p_i$, and the same in q_i; insert in (5.1). The product rule is the case $F(u, v) = uv$. ∎

By antisymmetry, linearity and the Leibniz rule hold in the first slot too. Fix f and write $D_f = \{f, \cdot\}$, that is,

$$D_f = \sum_{i=1}^{n} \Big(\frac{\partial f}{\partial q_i} \frac{\partial}{\partial p_i} - \frac{\partial f}{\partial p_i} \frac{\partial}{\partial q_i} \Big),$$

a *first-order differential operator*: an operator $D = \sum_k a_k(x)\, \partial/\partial x_k$ in the $2n$ variables x, with C^∞ coefficients. Leibniz says that D_f is a *derivation* of the product, that is, a linear map that obeys the product rule. The bracket is not associative. Instead, the Jacobi identity relates the three ways of taking nested brackets of f, g, and h.

Lemma 5.4 (Commutators of first-order operators)**.** *If $D = \sum_k a_k \partial_k$ and $E = \sum_k b_k \partial_k$ are first-order operators, then $[D, E] = DE - ED$ is the first-order operator*

$$[D, E] = \sum_k \big(D b_k - E a_k \big)\, \partial_k .$$

Proof. For $u \in C^\infty$, the product rule gives

$$DEu = \sum_{j,k} a_j \partial_j (b_k \partial_k u) = \sum_k (D b_k) \partial_k u + \sum_{j,k} a_j b_k\, \partial_j \partial_k u .$$

Subtract the same with a and b exchanged. The second-order terms $\sum_{j,k} (a_j b_k - b_j a_k) \partial_j \partial_k u$ cancel, because $\partial_j \partial_k u = \partial_k \partial_j u$. ∎

Theorem 5.5 (The Jacobi identity)**.** *For all observables f, g, h,*

$$\{f, \{g, h\}\} + \{g, \{h, f\}\} + \{h, \{f, g\}\} = 0. \tag{5.6}$$

Proof. Expand the left side by (5.1). The coefficients in (5.1) are constants, so each term of the expansion is a product of first derivatives of two of the three functions and one second derivative of the third. Collect the terms that contain a second derivative of h. The third bracket $\{h, \{f, g\}\}$ contains only first derivatives of h, so these terms all come from

$$\{f, \{g, h\}\} + \{g, \{h, f\}\} = \{f, \{g, h\}\} - \{g, \{f, h\}\} = (D_f D_g - D_g D_f) h,$$

and they are its second-order part. By Lemma 5.4 that part vanishes. The cyclic sum is unchanged when f, g, h are permuted cyclically, so the same argument removes the terms with a second derivative of f and those with a second derivative of g. No term is left. ∎

By antisymmetry, (5.6) can be rewritten in two useful forms:

$$\{f, \{g, h\}\} = \{\{f, g\}, h\} + \{g, \{f, h\}\}, \qquad \{\{f, g\}, H\} = \{\{f, H\}, g\} + \{f, \{g, H\}\}. \tag{5.7}$$

The first says that D_f is a derivation of the bracket, as Leibniz says it is a derivation of the product. The second says that the rate $\{\cdot, H\}$ obeys a product rule for brackets; it is the form used below.

Remark 5.6 (A non-constant bracket)**.** Only one step used the constant coefficients: that every term has a second derivative. For a bracket $\{f, g\} = \sum_{a,b} \Pi_{ab}(x)\, \partial_a f\, \partial_b g$ with a C^∞ antisymmetric matrix $\Pi(x)$, the second-order terms still cancel (each $\{f, \cdot\}$ is still first order), but terms with derivatives of Π remain. They are trilinear in first derivatives, so the cyclic sum equals $\sum_{a,b,c} \partial_a f\, \partial_b g\, \partial_c h$ times the cyclic sum for the coordinate functions x_a, x_b, x_c (whose second derivatives vanish): Jacobi holds for all observables if and only if it holds for the coordinates. So the identity need only be checked on the coordinate functions. Lecture 11 uses this, for a bracket whose matrix varies with the point.

Definition 5.7 (Poisson algebra)**.** A *Poisson algebra* is a real vector space $\mathcal{A}$ with a commutative, associative product with unit and a bilinear bracket $\{\,,\,\}$ that is antisymmetric, satisfies the Jacobi identity, and satisfies the Leibniz rule

$$\{f, gh\} = \{f, g\}\, h + g\, \{f, h\}\,.$$

Definition 5.8 (Lie algebra)**.** A *Lie algebra* is a real vector space with a bilinear, antisymmetric bracket that satisfies the Jacobi identity.

For example, $\mathbb{R}^3$ with the cross product is a Lie algebra. By Proposition 5.3 and Theorem 5.5, $C^\infty(\mathcal{M})$ with (5.1) is a Poisson algebra; as a Lie algebra it is infinite-dimensional, the *Poisson–Lie algebra of observables*. A Poisson algebra has two operations related by Leibniz. The product combines readings, and the bracket with H determines their rate of change in time.

Proposition 5.9 (The centre)**.** *Let $\mathcal{M}$ be connected, that is, any two of its points are joined by a path in $\mathcal{M}$. If $\{f, g\} = 0$ for every $g \in C^\infty(\mathcal{M})$, then f is constant.*

Proof. Take $g = p_i$ and $g = q_i$ in (5.4): $\partial f/\partial q_i = \partial f/\partial p_i = 0$ for every i. A function with zero gradient on a connected open set is constant. ∎

Conversely, constants commute with everything, so on a connected $\mathcal{M}$ the *centre* $\{f : \{f, \cdot\} = 0\}$ is $\mathbb{R} \cdot 1$. On a disconnected $\mathcal{M}$, a function that is constant on each component is also central. Lecture 11 meets brackets for which the centre is much larger.

5.4 Conserved quantities

Definition 5.10 (Conserved quantity)**.** A time-dependent observable f is *conserved* (an *integral of the motion*) if $f(x(t), t)$ is constant along every solution of Hamilton's equations in $\mathcal{M} \times I$.

Proposition 5.11 (Conservation is a bracket equation · *proved; "only if" uses T1 (cited)*)**.** *An observable f is conserved if and only if*

$$\frac{\partial f}{\partial t} + \{f, H\} = 0 \quad \text{on } \mathcal{M} \times I. \tag{5.8}$$

For f without explicit t this reads $\{f, H\} = 0$. If H does not depend on t, then H is conserved.

Proof. By (5.2) the derivative of $f(x(t), t)$ along a solution is the left side of (5.8) at $(x(t), t)$, so (5.8) implies conservation. Conversely, through every $(x_0, t_0) \in \mathcal{M} \times I$ passes a solution (Theorem 1.3, cited, applied to (1.6)); at t_0 the derivative of $f(x(t), t)$ is the left side at (x_0, t_0), and it vanishes. For $f = H$, $\{H, H\} = 0$. ∎

A conserved quantity can depend explicitly on time.

Example 5.12 (The boost)**.** For a free particle in $\mathbb{R}^3$, $H = |p|^2/2m$, take the vector $\mathbf{K} = mq - tp$, which depends on the time explicitly. Check (5.8) componentwise: $\partial_t K_i = -p_i$, and $\{K_i, H\} = m\,\{q_i, H\} = p_i$ by (5.5); the sum vanishes. Its value on the motion $q(t) = q_0 + tp/m$ is $\mathbf{K} = mq_0$: m times the initial position. Its brackets with the momenta are

$$\{K_i, p_j\} = m\,\delta_{ij}, \tag{5.9}$$

by (5.3).

Theorem 5.13 (Poisson · *proved; uses Proposition 5.11, hence T1 (cited)*)**.** *If f and g are conserved, so is $\{f, g\}$. The conserved quantities form a Lie subalgebra of the time-dependent observables, and they are also closed under products and smooth functions.*

Proof. The bracket is taken at fixed t, and f, g are C^∞, so $\partial_t \{f, g\} = \{\partial_t f, g\} + \{f, \partial_t g\}$. With the second form of (5.7),

$$\frac{\partial}{\partial t}\{f, g\} + \{\{f, g\}, H\} = \left\{\frac{\partial f}{\partial t} + \{f, H\},\ g\right\} + \left\{f,\ \frac{\partial g}{\partial t} + \{g, H\}\right\} = 0$$

by Proposition 5.11. Products and smooth functions of quantities that are constant along each motion are constant along each motion. ∎

Poisson's theorem may produce a new integral, or nothing new. For the boost, $\{K_i, p_j\} = m\delta_{ij}$ is a constant. It gives no additional information about the motion. For a central force all three components of $\mathbf{L} = q \times p$ are conserved (Lecture 3), and $\{L_x, L_y\} = L_z$ (§ 5.5) returns a quantity already known.

5.5 Angular momentum

Let $q, p \in \mathbb{R}^3$. Write ε_{abc} for the alternating symbol: it has $\varepsilon_{123} = 1$, and it changes sign when two indices are exchanged, so it vanishes when two indices coincide. Write $L_a = \sum_{j,k} \varepsilon_{ajk} q_j p_k$ for the components of $\mathbf{L} = q \times p$. A triple u of observables is a *vector* if $\{L_a, u_b\} = \sum_c \varepsilon_{abc} u_c$, and an observable s is a *scalar* if $\{L_a, s\} = 0$, for all a, b. For constant $n \in \mathbb{R}^3$, componentwise,

$$\{n \cdot \mathbf{L}, u\} = u \times n, \qquad \{n \cdot \mathbf{L}, s\} = 0. \tag{5.10}$$

By antisymmetry, $\{u, n \cdot \mathbf{L}\} = n \times u$. This is the rate at which a vector changes when it turns about a unit vector n at unit speed, and Lecture 6 shows that the flow of $n \cdot \mathbf{L}$ is this rotation. That is the reason for the word "vector".

Proposition 5.14 (**L** rotates vectors)**.** *The variables q and p are vectors. If u, w are vectors and $s, s_1, \dots, s_k$ are scalars, then $u \cdot w$ and $F(s_1, \dots, s_k)$ (F smooth on an open set containing the values of $(s_1, \dots, s_k)$) are scalars, and $u \times w$, $u + w$ and su are vectors.*

Proof. First, q and p are vectors. By Leibniz and (5.3),

$$\{L_a, q_b\} = \sum_{j,k} \varepsilon_{ajk} q_j \{p_k, q_b\} = -\sum_j \varepsilon_{ajb} q_j = \sum_c \varepsilon_{abc} q_c,$$
$$\{L_a, p_b\} = \sum_{j,k} \varepsilon_{ajk} \{q_j, p_b\} p_k = \sum_c \varepsilon_{abc} p_c.$$

Next, $u \cdot w$ is a scalar. By Leibniz and (5.10),

$$\{n \cdot \mathbf{L}, u \cdot w\} = (u \times n) \cdot w + u \cdot (w \times n) = 0,$$

since the two triple products equal $\pm n \cdot (w \times u)$ with opposite signs. Finally, $u \times w$ is a vector, since

$$\{n \cdot \mathbf{L}, u \times w\} = (u \times n) \times w + u \times (w \times n) = (u \times w) \times n$$

by the Jacobi identity of the cross product. The rest is linearity and the composition rule of Proposition 5.3. ∎

Hence $\mathbf{L} = q \times p$ is a vector:

$$\{L_a, L_b\} = \sum_c \varepsilon_{abc} L_c, \qquad \text{e.g. } \{L_x, L_y\} = L_z. \tag{5.11}$$

This last bracket can be checked directly. With $L_x = q_y p_z - q_z p_y$ and $L_y = q_z p_x - q_x p_z$, Leibniz and (5.3) leave only two terms:

$$\{L_x, L_y\} = \{q_y p_z, q_z p_x\} + \{q_z p_y, q_x p_z\} = -q_y p_x + q_x p_y = L_z.$$

By Poisson's theorem, *if two components of* $\mathbf{L}$ *are conserved, so is the third*. The scalars include $|q|^2$, $|p|^2$, $q \cdot p$, $|\mathbf{L}|^2$, and, on $q \neq 0$, every smooth function of $|q|$. So every $H = |p|^2/2m + V(|q|)$ with $V \in C^\infty(0,\infty)$ is a scalar, and all three L_a are conserved, by Proposition 5.11.

5.6 The Heisenberg picture

So far we evaluated a fixed observable along a moving state. Now let H not depend on t, with a complete flow φ_t (Definition 1.9), as for $\frac{1}{2}|p|^2 + V(q)$ with $V \in C^\infty(\mathbb{R}^n)$ bounded below (Proposition 1.10). The *Heisenberg picture* keeps the state fixed and lets the observable change: $f_t := f \circ \varphi_t$. Thus $f_t(x)$ is the reading of f at time t on the motion that starts at x.

Theorem 5.15 (Theorem A: the Heisenberg equation · *proved modulo T2 (cited)*)**.** *Let* $f \in C^\infty(\mathcal{M})$.

1. $(t,x) \mapsto f_t(x)$ *is* C^∞*, and*
$$\frac{\partial f_t}{\partial t} = \{f_t, H\} = \{f, H\} \circ \varphi_t. \tag{5.12}$$
2. *(Uniqueness) If* $F\colon \mathbb{R} \times \mathcal{M} \to \mathbb{R}$*,* $(t,x) \mapsto F_t(x)$*, is* C^1*, with* $\partial_t F_t = \{F_t, H\}$ *and* $F_0 = f$*, then* $F_t = f_t$ *for all* t*.*

Proof. (1) The flow is C^∞ in (t,x) (Theorem 1.6), and by the group law (1.7) $f_{t+s} = f_t \circ \varphi_s = f_s \circ \varphi_t$. Differentiate at $s = 0$: the curve $s \mapsto \varphi_s(y)$ has velocity $J\nabla H(y)$ at $s = 0$, so the chain rule turns the two expressions into $\nabla f_t \cdot J\nabla H(x) = \{f_t, H\}(x)$ and $\{f, H\}(\varphi_t x)$. Both equal $\partial_t f_t(x)$.

(2) Fix t and x, and let $G(s) = F_{t-s}(\varphi_s(x))$ for s between 0 and t. By the chain rule,

$$G'(s) = -\big(\partial_t F\big)_{t-s}(\varphi_s x) + \{F_{t-s}, H\}(\varphi_s x) = 0.$$

So $F_t(x) = G(0) = G(t) = F_0(\varphi_t x) = f_t(x)$. ∎

The theorem separates two meanings of "$\mathrm{d}f/\mathrm{d}t$". Along the motion $x(t) = \varphi_t x$ the *number* $f(x(t))$ changes at the rate $\{f, H\}(x(t))$ (5.2); in the Heisenberg picture the *function* f_t changes at the fixed point x at the rate $\{f_t, H\}(x)$. By (5.12) the two rates are equal: one may first evolve the observable and then bracket it with H, or take its bracket with H first and then evolve it. For two observables f and g the corresponding statement is $\{f,g\} \circ \varphi_t = \{f \circ \varphi_t, g \circ \varphi_t\}$. This is Theorem B (Theorem 6.4), and it rests on Jacobi. The level sets of f_t are those of f carried *backwards*: $\{f_t = c\} = \varphi_{-t}(\{f = c\})$ (Figure 5.3).

Example 5.16 (The oscillator)**.** Let $H = p^2/2m + \frac{1}{2}m\nu^2 q^2$; its flow is complete, since Hamilton's equations are linear. Its observables q_t and p_t are

$$q_t = q\cos\nu t + \frac{p}{m\nu}\sin\nu t, \qquad p_t = p\cos\nu t - m\nu q \sin\nu t,$$

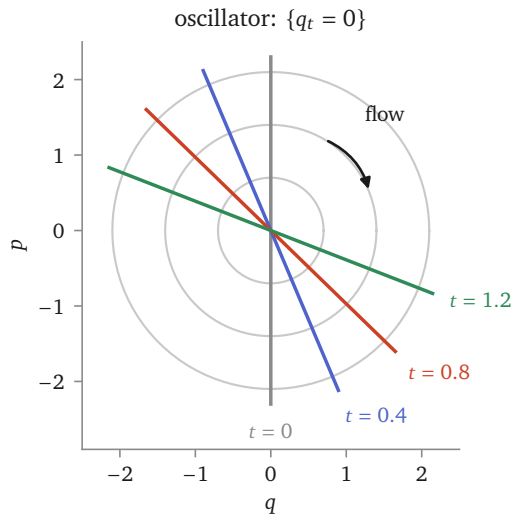


Figure 5.3: The set $\{q_t = 0\} = \varphi_{-t}(\{q = 0\})$ of states that reach $q = 0$ after time t, for the oscillator, $m = \nu = 1$: states turn clockwise, the line $q\cos t + p\sin t = 0$ counter-clockwise.

for every t. Each right side is C^∞ and equals q (resp. p) at $t = 0$. Each also solves $\partial_t F_t = \{F_t, H\}$. For the first,

$$\{F_t, H\} = \cos \nu t \, \frac{p}{m} - \frac{\sin \nu t}{m\nu} \, m\nu^2 q = \frac{p}{m} \cos \nu t - \nu q \sin \nu t = \partial_t F_t,$$

and the second is the same computation. So the uniqueness statement of Theorem 5.15 identifies the right sides with q_t and p_t. The state $(1, 0)$ moves to $(\cos \nu t, -m\nu \sin \nu t)$: clockwise, the sign fact that § 6.1 records.

5.7 Finite-dimensional subalgebras

The algebra $C^\infty(\mathcal{M})$ is infinite-dimensional, but polynomials in (q, p) contain finite-dimensional Lie algebras. We find them by counting degrees.

Proposition 5.17 (Degree counting). *If f and g are polynomials in (q, p) of degrees a and b, then $\{f, g\}$ is a polynomial of degree at most $a + b - 2$. If f and g are homogeneous, $\{f, g\}$ is homogeneous of degree $a + b - 2$, or zero.*

Proof. Each term of (5.1) is a product of a derivative of f, of degree at most $a - 1$, and a derivative of g, of degree at most $b - 1$. ∎

So the polynomials of degree at most 2 form a Lie subalgebra, and inside it:

- *The Heisenberg algebra* $\mathfrak{h}_n = \operatorname{span}\{1, q_1, \dots, q_n, p_1, \dots, p_n\}$, of dimension $2n + 1$, with

$$\{q_i, p_j\} = \delta_{ij}\, 1, \qquad \{q_i, q_j\} = \{p_i, p_j\} = 0, \qquad \{1, \cdot\} = 0. \tag{5.13}$$

  Its centre is $\mathbb{R} \cdot 1$, and the constant 1 appears as a bracket.
- *The homogeneous quadratics*, of dimension $n(2n + 1)$: closed, since $2 + 2 - 2 = 2$. For $n = 1$, a basis is qp, $p^2/2$, $q^2/2$, with

$$\{qp, p^2/2\} = p^2, \qquad \{qp, q^2/2\} = -q^2, \qquad \{q^2/2, p^2/2\} = qp.$$

A quadratic bracketed with a linear function is linear ($2+1-2=1$), so the quadratics act on $\operatorname{span}\{q_i, p_i\}$ by linear maps. Lecture 6 identifies this Lie algebra with $\mathfrak{sp}(2n)$.

- $\mathfrak{so}(3)$*:* for $n=3$ the components L_a are homogeneous quadratics, and (5.11) says that their span is a subalgebra with the bracket of $(\mathbb{R}^3, \times)$.

Degree 3 is not closed: $\{q^3, p^3\} = 9q^2p^2$, and repeated brackets increase the degree. The polynomials of degree at most 2 are where classical and quantum brackets can be matched exactly (Lecture 12).

Summary.

- *Evolution:* observables and the bracket, with $\mathrm{d}f/\mathrm{d}t = \{f, H\} + \partial_t f$ along motions (Definition 5.2). For H without t and a complete flow, Theorem A says that $f_t = f \circ \varphi_t$ satisfies $\partial_t f_t = \{f_t, H\}$ (Theorem 5.15).
- *Algebra:* the Jacobi identity makes $C^\infty(\mathcal{M})$ a Poisson algebra (Theorem 5.5).
- *Conservation:* conservation is a bracket equation, and conserved quantities form a Lie algebra (Proposition 5.11 and Theorem 5.13).
- *Subalgebras:* angular momentum, with $\{L_a, L_b\} = \sum_c \varepsilon_{abc} L_c$; the Heisenberg algebra and the quadratics (Proposition 5.17).
- *Next:* Lecture 6 asks what flow each observable generates; Jacobi will say that these flows preserve the bracket.

References

Faddeev and Yakubovskii, *Lectures on Quantum Mechanics for Mathematics Students*, §1 (pp. 1–6), and Arnold, *Mathematical Methods of Classical Mechanics*, §40 (pp. 214–217): the proof of Jacobi by counting second derivatives is theirs. Takhtajan, *Quantum Mechanics for Mathematicians*, Ch. 1, §2.4 (pp. 38–39) and §2.6 (pp. 46–47). Tong, *Lectures on Classical Dynamics*, §4.3 (pp. 94–96).